

Fluid Flow in Pipes

Pre-Lab Plan

Unit Operations Lab 5

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Group #2, Tuesday, PM section

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1. Experimental Objective (5pts)

The objective of this experiment is to compare experimentally acquired data from a fluid flow system to their theoretical values. This data, used to calculate energy loss coefficients like friction factor (f), loss coefficient (K), and flow coefficient (C_f), are compared to such values like the Fanning friction and Darcy friction factor. The parts of the fluid flow system being measured include an orifice meter, an expansion and contraction pipe, and straight pipes made of plastic, copper, and stainless steel, all at different diameters.

2. Experimental

The apparatus and also the materials and supply of this experiment are introduced below and briefly explained their usage.

2.1 Apparatus (5 pts)



Figure 1: Apparatus used during Fluid Flow in Pipes experiment.

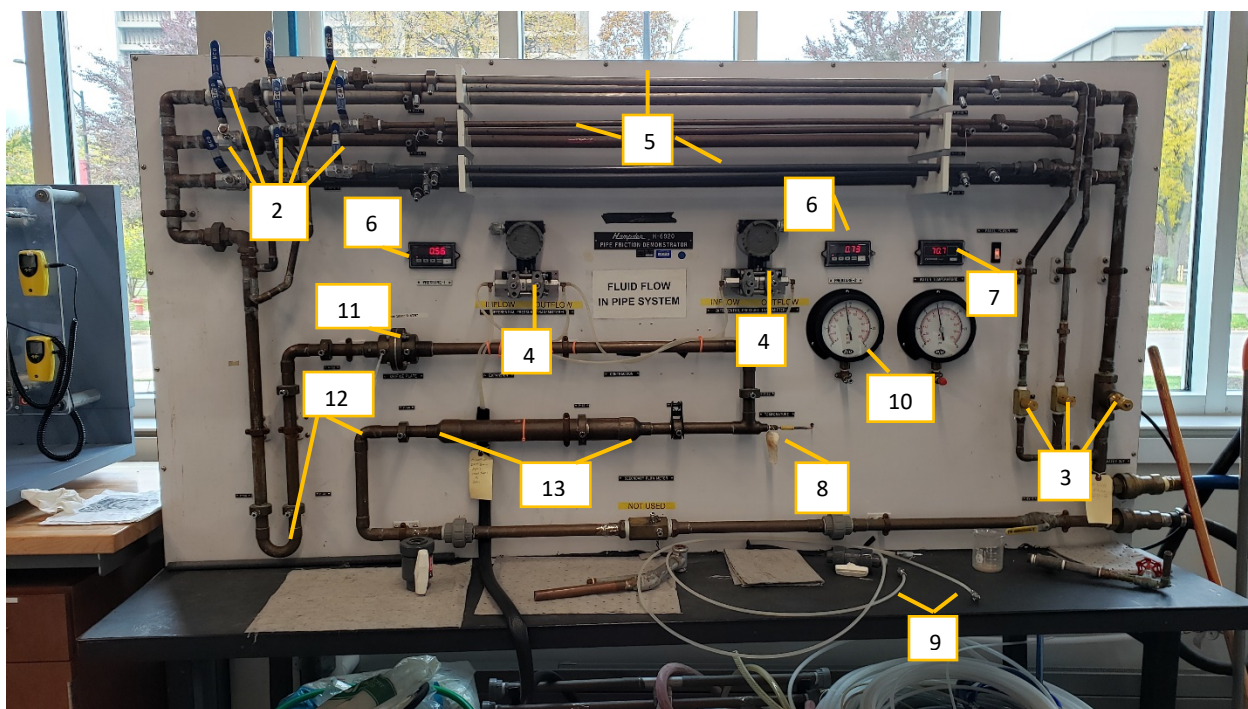


Figure 2: Apparatus used during Fluid Flow in Pipes experiment.

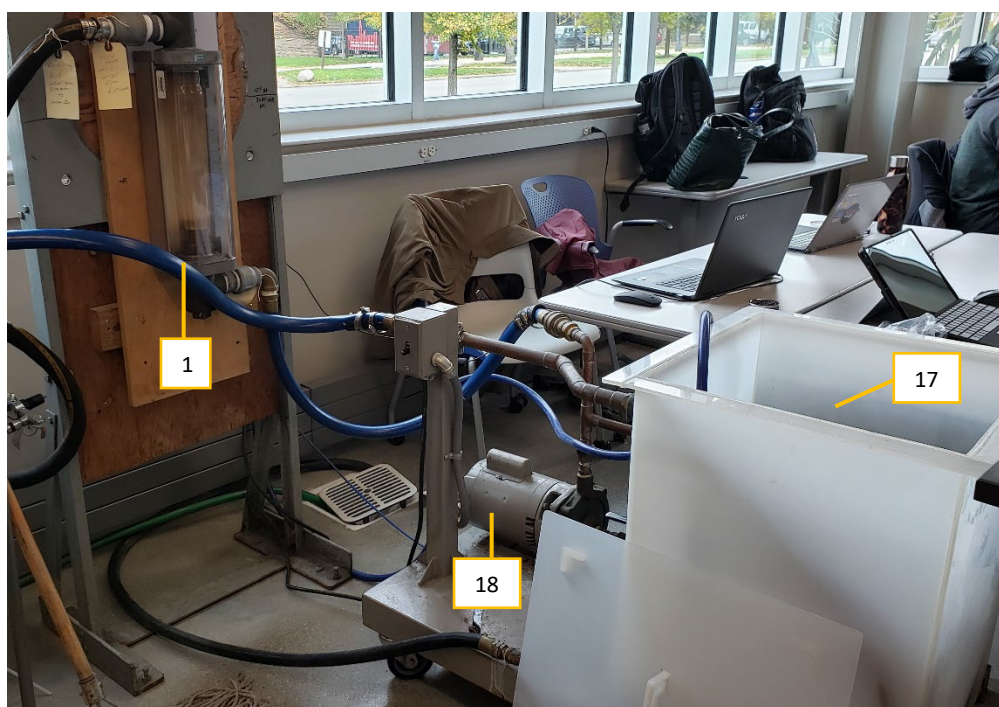


Figure 3: Apparatus used during Fluid Flow in Pipes experiment.

Table 1. List of apparatus in the filtration unit with their description

1	Rotameter	Used to measure fluid flowrate in the piping
2	Shut-Off Valves	Valves used to open and close section of piping system
3	Flow Control Valves	Valves used to regulate flow rate through piping system
4	Pressure Taps	Used to measure pressure
5	Flow Circuit	3 Copper, 3 Stainless Steel, 3 Plastic straight pipes of 3 different diameters
6	Differential Pressure Transmitter	Digital device used to measure pressure drop in psig
7	Temperature Monitor	Digital device used to measure the fluid temperature in piping system, in °F units.
8	Thermocouple	Device attached to pipes designed for temperature readings.
9	Quick Fittings	Plastic tubing used for quick attaching to pressure taps in order to measure pressure.
10	Analog Pressure Gauge (2)	Device used to measure pressure in psi or kPa
11	Orifice Plate	Special fitting use to attach two sections of piping. (D=0.478")
12	90/180° Bend	Special fittings use to connect piping with 90° and 180° angles.
13	Expansion/Contraction Fittings	Special fittings use to connect piping with different diameters.
14	Short Radius Elbow	Special fitting with a tight 90° angle
15	Secondary Flow Meter	Special fitting used to measure flowrate
16	Rubber Tubing	Tubing used to transport fluid between reservoir tank and piping system.
17	Reservoir Tank	Water container for pumping through piping system.
18	Pump	Centrifugal pump used to drive flow of fluid from reservoir tank to the piping system.

2.2 Materials and Supplies (5 pts)

Table 2: Description of materials and supplies used during this experiment.

1	Water	Tap water at room temperature
2	Graduated Bucket	Container used to measure volume
3	Rag	Used to clean-up spills.
4	Bucket	Container used to collect fluids.
5	Tubes	Tubes used to connect pressure taps to DPT

2.3 Experimental Plan (10 pts)

- **Set up**

- Make sure the pressure transmitters are calibrated. If readings are not close to zero (0.00 to 0.02), communicate it to the TA or instructor.
- When going from one measurement to another, close all the valves that leads to water flow. Disconnect the quick connection fitting at the sensing ports and connect them to the next section being measured. Upstream pressure sensing port should be connected to the high side of the DPT, while downstream should be connected in the lower side. The difference in pressure displayed by the digital meter is a direct reading of pressure drop in psi.
- Open the valves back up.
- To flush out the DPT, just open the two black valves on the DPT's side, after flushing, close the valves.
- After the flow in the piping become steady, take measurements of the pressure differential directly from the DPT meter.

- **Procedure**

- To calibrate the rotameter, measure the flow rate with a graduated cylinder.
- Determine experimentally the conversion factor and compare it with the theoretical data in LPM units.
- Use the orifice meter to measure pressure drop, flow rate, temperature, and to determine the flow coefficient C_f experimentally. Observe if C_f varies with the change in flow rate or if it remains constant.
- For the pipe friction head loss, measure the pressure differential and experimentally determine friction coefficient. Compare the data obtained with theoretical data. Perform the same measurements for copper, stainless steel and plastic tubing.
- For fitting head loss, measure the pressure differential, the flow rate and temperature. Determine the loss coefficient K experimentally.
- Repeat the procedure two times, plot the results and determine the accuracy of your data by doing a best fit curve and by calculating R^2 .

Project Safety Evaluation (10 pts)

Project Title: Fluid Flow in Pipes

Date: 10/29/2019

	Yes/No	
Potential electrical hazards?	yes	If yes, identify conditions: Possibility of electrical shock due to electrical switch and water.
Potential mechanical hazards?	yes	If yes, identify conditions: Centrifugal pump rotating shaft; make sure to not run the pump without water.
Potential pressure hazards?	yes	If yes, identify conditions: Pump has an outlet pressure of ~ 60 psi, don't close all valves in system, pressure can build up and damage the system.
Potential temperature hazards?	no	If yes, identify conditions:
Hazardous/toxic chemicals involved?	no	If yes, hazards involved:
Gloves required?	no	If yes, specify type:
MSDS reviewed by all team members?	yes	List MSDS sheets reviewed: Water - ACC# 00199 (Fischer scientific)
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	25 °C	No temperature hazards, water is at room temperature.
Pressure (psia)	60 psia	You must allow fluid to flow through the system when pump is on, do not close all the valves when the pump is on.
Safety Controls	yes/no	If yes, Provide description
Emergency Shut-off Switch and Valves	yes	Shut off valves and power switches in case of emergency.
Pump Switch	yes	Shuts off the power to the pump.

I have read relevant background material for the Unit Operations Laboratory entitled: “Fluid Flow in Pipes” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

James Fell 10/29/2019

signature of team member date

Jasmine Villegas 10/29/2019

signature of team member date

Mariana St. Peter 10/29/2019

signature of team member date

William Rawson 10/29/2019

signature of team member date